

Lab: Epistemological Analysis of the Free Energy Principle

Learning Goal: Apply philosophical analysis tools to the FEP's epistemological claims.

Part 1: Mapping the Epistemological Tradition

Exercise: Complete the mapping table:

Philosophical Concept	FEP Analog	Key Similarity	Key Difference
Kant's categories	Generative model		
Justified true belief	Model evidence		
Cartesian doubt	Prior uncertainty		
Empiricist sensation	Sensory states		
Pragmatist truth	Predictive success		

Write a 200-word analysis of the strongest and weakest mappings.

Write your response here

Part 2: The Markov Blanket Debate

Learning Goal: Evaluate competing interpretations of MB ontology.

Exercise: Read the following two positions and argue for each:

Position A (Realist): "Markov Blankets are real statistical structures that exist independently of our analysis. A cell membrane is a Markov Blanket whether or not anyone describes it as such."

Position B (Constructivist): "Markov Blankets are modeling tools imposed by the analyst. The same system can be partitioned differently depending on the question being asked."

Write 200 words defending Position A, then 200 words defending Position B. Finally, state your own position with reasons.

Write your response here

Part 3: Circularity Analysis

Learning Goal: Diagnose and evaluate the self-evidencing circularity.

Exercise:

1. Formalize the circularity: Model M defines what counts as evidence E. E is used to update M. There is no external check.
2. Is this vicious circularity (fatal to the framework) or virtuous circularity (a feature)?
3. Compare with at least two other cases of circularity in philosophy or science (e.g., the hermeneutic circle, defining health in terms of function and function in terms of health).

Write your response here

Part 4: Is the FEP Falsifiable?

Learning Goal: Apply philosophy of science criteria to the FEP.

Exercise: Apply Popper's falsifiability criterion:

1. State the FEP as precisely as possible.
2. What observation would *falsify* it?
3. If it cannot be falsified, is it a tautology? A framework? A research programme (in Lakatos's sense)?
4. Compare with other unfalsifiable-seeming frameworks (e.g., natural selection, thermodynamics).

Write your response here

Part 5: Reflection

In 250 words, reflect: Does the FEP provide genuine epistemological insight, or does it merely translate existing philosophical ideas into mathematical notation? What is *new* that philosophy couldn't already offer?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Historical mapping	FEP in philosophical tradition
2	Dialectical argument	Markov Blanket ontology
3	Circularity analysis	Self-evidencing
4	Philosophy of science	Falsifiability
5	Critical reflection	Novelty of the FEP